

SECURITY ENGINEERING — HANDS-ON LAB

Automated Sign-In Risk Detection & Response

Microsoft Entra ID → Microsoft Sentinel → Logic Apps

David Rufino Botte | Senior Desktop Engineer

github.com/davidrufinobotte/entra-sentinel-detection-lab

HOW IT WORKS

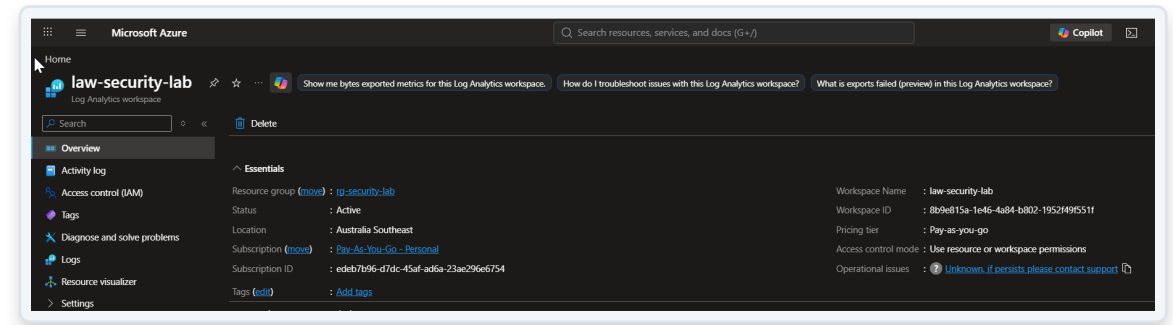
End-to-end pipeline: log to alert



A risky sign-in is logged, ingested into Sentinel, evaluated by a scheduled detection rule, and — if it matches — automatically opens an incident and triggers an emailed alert.

Log Analytics Workspace

- Created law-security-lab as the central log repository for the lab
- Pay-As-You-Go subscription, Australia Southeast region
- Base layer every downstream Sentinel component reads from



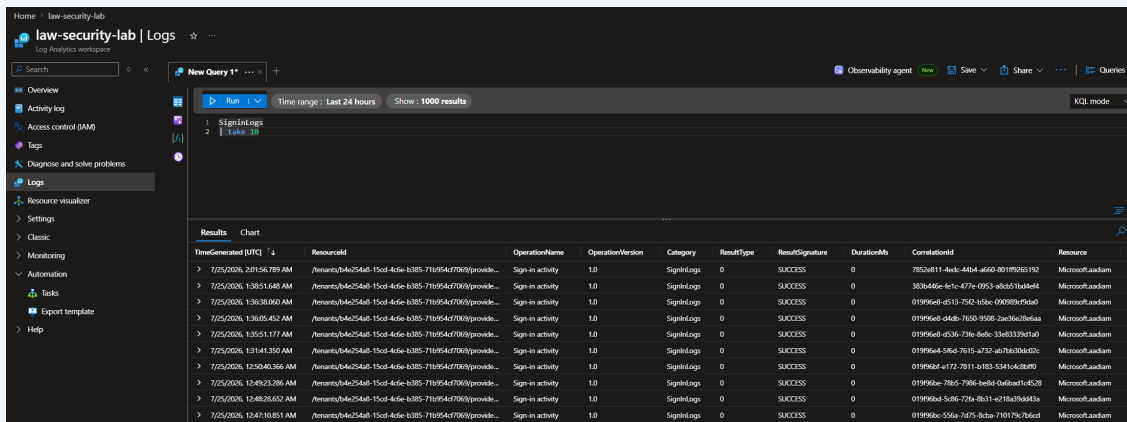
Connecting Identity Signals

- Connected the Microsoft Entra ID data connector (Sign-In + Audit logs)
- Fixed a real issue: the connector was missing until the Entra ID solution was installed from the Sentinel Content Hub
- Confirms troubleshooting ability, not just following steps

The screenshot displays the Microsoft Defender console interface. On the left, the navigation pane shows 'Data connectors' under the 'Analytics' section, with 'Microsoft Sentinel' and 'Data connectors' highlighted. The main pane shows the 'Connector details' for 'Microsoft Entra ID'. The 'Connected Status' is 'Provider', and the 'Last Log Received' is '3 Hours Ago'. The 'Description' states: 'Gain insights into Microsoft Entra ID by connecting Audit and Sign-in logs to Microsoft Sentinel to gather insights around Microsoft Entra ID scenarios. You can learn about app usage, conditional access policies, legacy auth relate details using our Sign-in logs. You can get information on your Self Service Password Reset (SSPR) usage. Microsoft Entra ID Management activities like user, group, role, app management using our Audit logs table.' The 'Last data received' is '26/07/2026, 09:23:24'. The 'Content source' is 'Microsoft Entra ID' and the 'Version' is '1.0.0'. The 'Author' is 'Microsoft' and it is 'Supported by Microsoft Corporation'. The 'Related content' shows '0 Workbooks', '2 Queries', and '73 Analytics rules templates'. A line graph shows 'Data received' over time from 19 July to 25 July. The 'Table management' section shows two tables: 'AuditLogs' and 'SignInLogs', both with 'Analytics' tier, 'Sentinel' table type, '30 days' analytics retention, and '30 days' total retention. The 'Workspace' is 'law-security-lab'. The 'Configuration' section shows 'Connect Microsoft Entra ID logs to Microsoft Sentinel' with 'Sign-In Logs' and 'Audit Logs' selected. A note states: 'In order to export Sign-in data, your organization needs Microsoft Entra ID P1 or P2 license. If you don't have a P1 or P2, start a free trial.' Below this, there are checkboxes for 'Non-Interactive User Sign-In Log', 'Service Principal Sign-In Logs', and 'Anonymous Activity Sign-In Logs'.

Confirming Data Ingestion

- Ran a KQL query (SignInLogs | take 10) before building any detection logic
- Verified logs were flowing end-to-end from Entra ID into the workspace
- Build on confirmed data, not assumptions



The screenshot shows the Microsoft Sentinel Log Analytics workspace interface. The left sidebar contains navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Logs (selected), Resource visualizer, Settings, Classic, Monitoring, Automation, Tasks, and Help. The main pane displays a KQL query named 'New Query 1' with the following query:

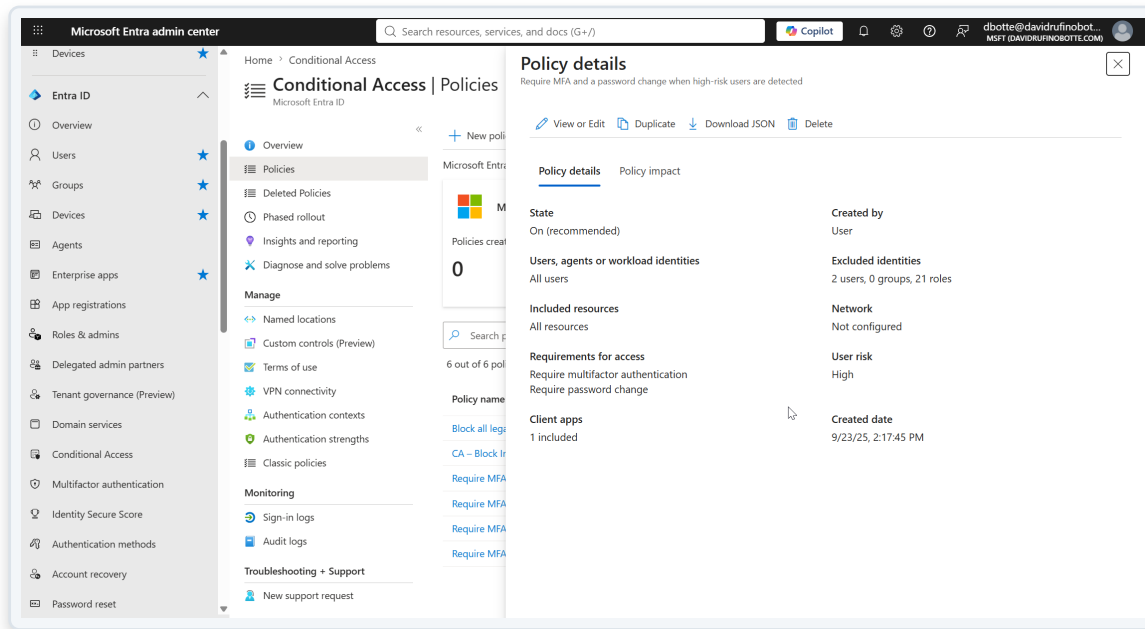
```
1 SignInLogs
2 | take 10
```

The query results are displayed in a table with the following columns: TimeGenerated (UTC), ResourceId, OperationName, OperationVersion, Category, ResultType, ResultSignature, DurationMs, CorrelationId, and Resource. The results show 10 log entries for 'Sign-in activity' from the resource 'ResourceId'.

TimeGenerated (UTC)	ResourceId	OperationName	OperationVersion	Category	ResultType	ResultSignature	DurationMs	CorrelationId	Resource
7/25/2024, 2:01:55:789 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	7832d811-4a4d-4804-a659-b0180351192	Microsoft.Laam
7/25/2024, 1:26:51:648 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	383d4d6a-8e-fc-477d-0953-d8d511d4d4d	Microsoft.Laam
7/25/2024, 1:26:30:000 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	01995d4d-c513-7542-83d6-090909191d4	Microsoft.Laam
7/25/2024, 1:26:05:403 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	01995d4d-c513-7542-83d6-090909191d4	Microsoft.Laam
7/25/2024, 1:25:51:177 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	01995d4d-c513-7542-83d6-090909191d4	Microsoft.Laam
7/25/2024, 1:25:40:286 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	01995d4d-c513-7542-83d6-090909191d4	Microsoft.Laam
7/25/2024, 12:50:40:286 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	01995d4d-c513-7542-83d6-090909191d4	Microsoft.Laam
7/25/2024, 12:49:23:286 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	01995d4d-c513-7542-83d6-090909191d4	Microsoft.Laam
7/25/2024, 12:48:20:002 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	01995d4d-c513-7542-83d6-090909191d4	Microsoft.Laam
7/25/2024, 12:47:10:001 AM	ResourceId	Sign-in activity	1.0	SignInLog	0	SUCCESS	0	01995d4d-c513-7542-83d6-090909191d4	Microsoft.Laam

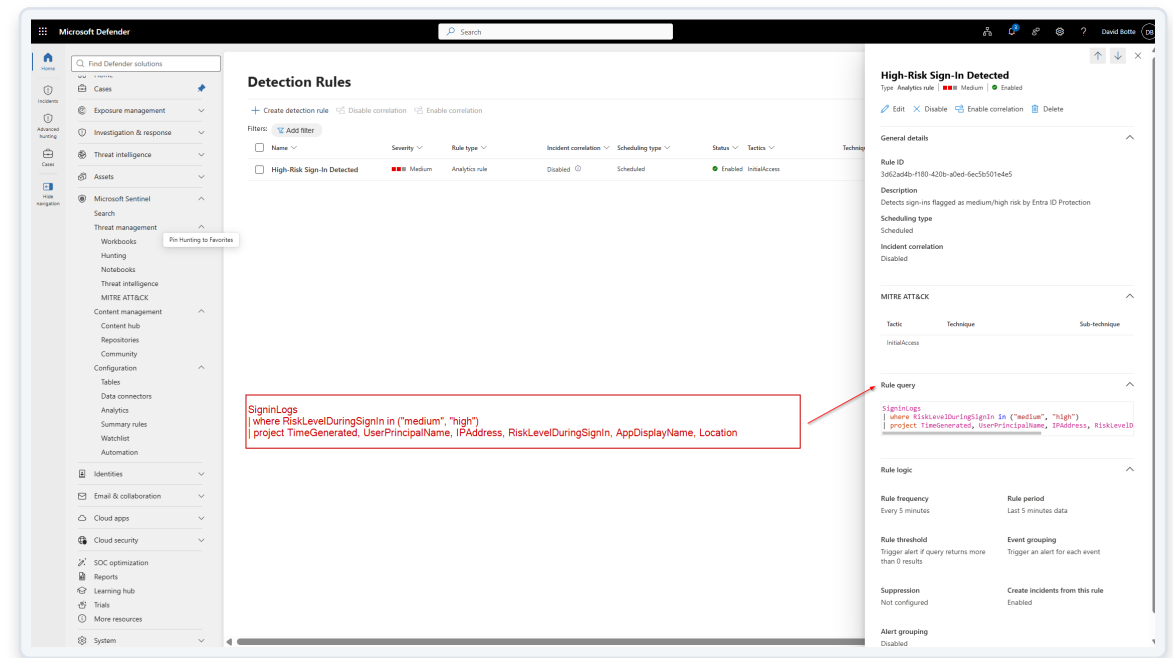
Reusing Existing Controls

- Audited existing Conditional Access policies before creating a new one
- Found and reused a policy already enforcing MFA on risky sign-ins
- Avoids policy sprawl — a real security engineering judgment call



Building the Analytics Rule

- Scheduled KQL rule 'High-Risk Sign-In Detected', running every 5 minutes (5 minutes just because is test enviroment)
- Filters on RiskLevelDuringSignIn (medium / high) from Entra ID Protection
- Mapped to MITRE ATT&CK — Initial Access



Detections at Scale

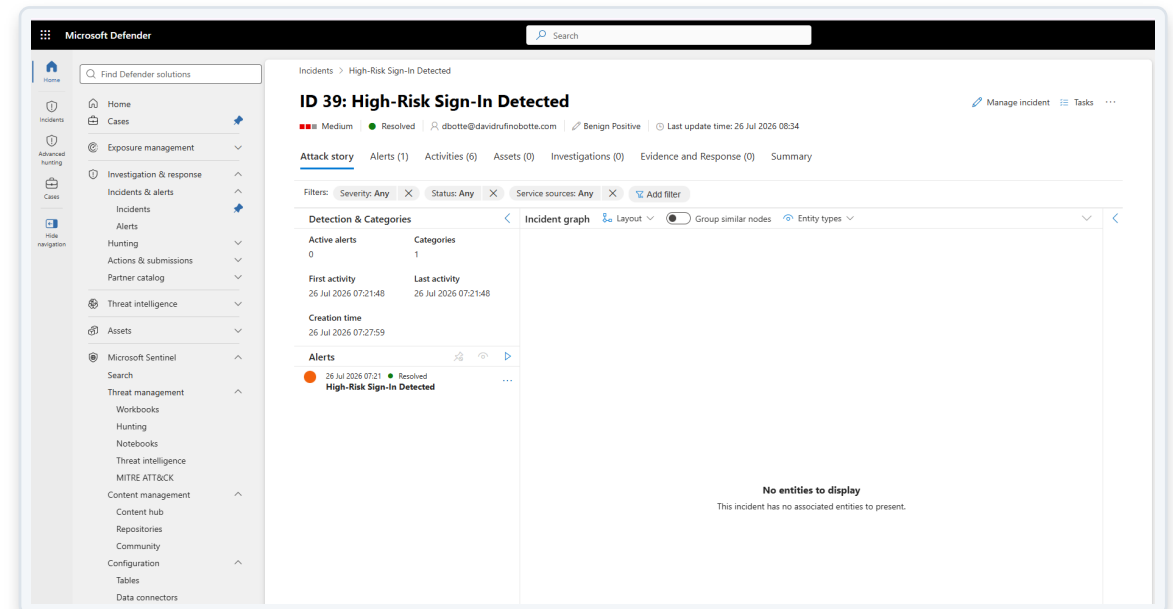
- The rule generated 39 incidents from real sign-in activity
- Confirms the detection logic works against live data, not just theory
- Each incident automatically linked back to its source alert

The screenshot displays the Microsoft Defender interface. On the left is a navigation pane with categories like Home, Cases, Investigation & response, Threat intelligence, Assets, and Microsoft Sentinel. The main area is titled 'Incidents' and shows a list of 39 incidents. A summary card indicates '0% Last 30 days' for multi-alert incidents. The incident list includes columns for Incident name, Incident ID, Priority, Severity, Investigation state, Categories, Impacted assets, Active alerts, and Service sources. All incidents are 'High-Risk Sign-In Detected' with a severity of 'Medium' and an investigation state of 'Initial access'.

Incident name	Incident ID	Priority	Severity	Investigation state	Categories	Impacted assets	Active alerts	Service sources
> High-Risk Sign-In Detected	39	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	38	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	37	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	35	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	34	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	36	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	33	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	31	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	32	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	30	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	29	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	28	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	27	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	26	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	25	2	Medium	Initial access			0/1	Microsoft Sentinel

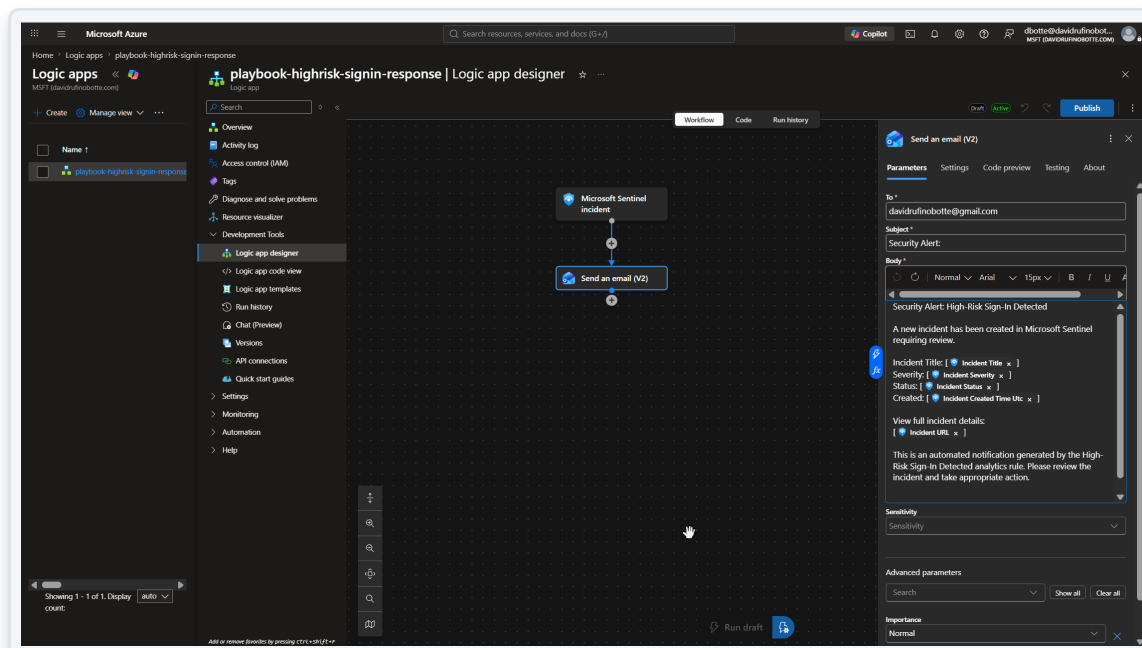
Incident Review & Closure

- Reviewed and closed incidents individually with correct classification
- Classified as Benign Positive — the detection was correct; the activity was a controlled test
- Standard SOC discipline, not blanket dismissal



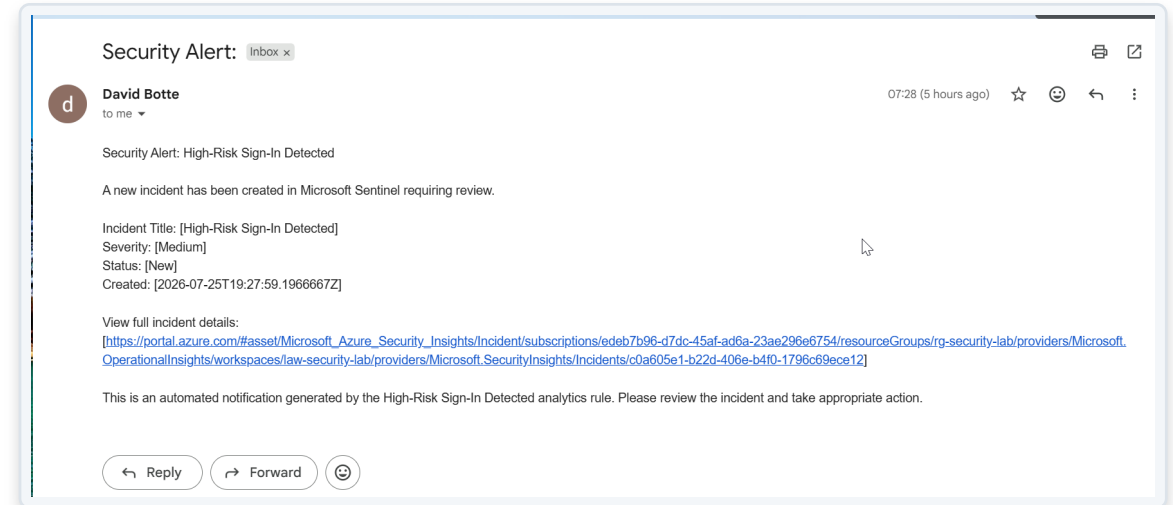
Automated Response Playbook

- Logic App triggered directly by 'Microsoft Sentinel incident'
- Sends a structured alert with incident title, severity, status and link
- Consumption plan used deliberately, to avoid fixed hosting cost



Closing the Loop

- Alert delivered by email within minutes of incident creation
- Full pipeline proven: sign-in log → detection → incident → automated notification
- No manual step between detection and alert delivery



WHAT THIS DEMONSTRATES

Skills applied in this lab

- **Identity Security**

Entra ID, Conditional Access, risk-based sign-in policy

- **Security Automation**

Logic Apps / SOAR-style automated incident response

- **Troubleshooting**

Diagnosed and resolved real platform issues (missing connector, permissions, cross-tenant setup)

- **SIEM Engineering**

Microsoft Sentinel, KQL detection rules, MITRE ATT&CK mapping

- **SOC Discipline**

Incident triage, classification, and closure — not just detection

- **Cost & Governance Awareness**

Reused existing controls; chose consumption pricing to avoid unnecessary cost